In [29]:
```
import pandas as pd
```

In [30]:
```
dataset=pd.read_csv('insurance_pre.csv')
```

In [31]:
```
dataset
```

Out[31]:

	age	sex	bmi	children	smoker	charges
0	19	female	27.900	0	yes	16884.92400
1	18	male	33.770	1	no	1725.55230
2	28	male	33.000	3	no	4449.46200
3	33	male	22.705	0	no	21984.47061
4	32	male	28.880	0	no	3866.85520
...	...	...	...	...	...	...
1333	50	male	30.970	3	no	10600.54830
1334	18	female	31.920	0	no	2205.98080
1335	18	female	36.850	0	no	1629.83350
1336	21	female	25.800	0	no	2007.94500
1337	61	female	29.070	0	yes	29141.36030

1338 rows × 6 columns

In [32]:
```
dataset=pd.get_dummies(dataset,drop_first=True)
```

In [33]:
```
dataset
```

Out[33]:

	age	bmi	children	charges	sex_male	smoker_yes
0	19	27.900	0	16884.92400	0	1
1	18	33.770	1	1725.55230	1	0
2	28	33.000	3	4449.46200	1	0
3	33	22.705	0	21984.47061	1	0
4	32	28.880	0	3866.85520	1	0
...	...	...	...	...	...	...
1333	50	30.970	3	10600.54830	1	0
1334	18	31.920	0	2205.98080	0	0
1335	18	36.850	0	1629.83350	0	0
1336	21	25.800	0	2007.94500	0	0
1337	61	29.070	0	29141.36030	0	1

1338 rows × 6 columns

In [34]:
```
dataset.columns
```
Out[34]:
```
Index(['age', 'bmi', 'children', 'charges', 'sex_male', 'smoker_yes'], dtype
='object')
```

In [35]:
```
independent=dataset[['age', 'bmi', 'children','sex_male', 'smoker_yes']]
```

In [36]:
```
independent
```
Out[36]:

	age	bmi	children	sex_male	smoker_yes
0	19	27.900	0	0	1
1	18	33.770	1	1	0
2	28	33.000	3	1	0
3	33	22.705	0	1	0
4	32	28.880	0	1	0
...	...	...	...	...	...
1333	50	30.970	3	1	0
1334	18	31.920	0	0	0
1335	18	36.850	0	0	0
1336	21	25.800	0	0	0
1337	61	29.070	0	0	1

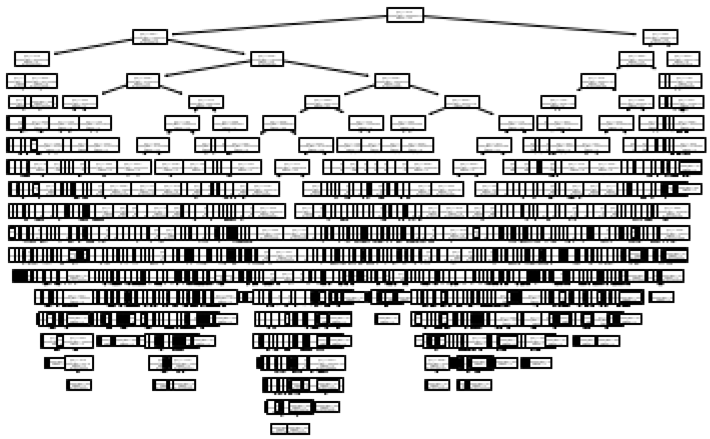
1338 rows × 5 columns

In [37]:
```
dependent=dataset[['charges']]
```

In [38]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(independent,dependent,test_size
=0.30,random_state=0)
```

In [77]:
```
from sklearn.tree import DecisionTreeRegressor
regressor=DecisionTreeRegressor(criterion='friedman_mse',splitter='random')
regressor=regressor.fit(x_train,y_train)
```

In [78]:
```
import matplotlib.pyplot as plt
from sklearn import tree
tree.plot_tree(regressor)
plt.show()
```



In [82]:
```
y_pred=regressor.predict(x_test)
```

In [83]:
```
from sklearn.metrics import r2_score
r_score=r2_score(y_test,y_pred)
```

In [84]:
```
r_score
```

Out[84]: 0.7001690051305376

In [ ]: